

Aryana Haghjoo

Curriculum Vitae

Department of Physics and Astronomy, University of California, Riverside, CA 92521, USA

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Research Interests

Machine learning as a measurement tool: calibrated spectral super-resolution, simulation-based inference, uncertainty quantification and failure detection · Galaxy evolution and emission-line diagnostics with JWST, Roman and Euclid · Reionization and cosmic dawn ($\text{Ly}\alpha$ damping wings, the 21 cm signal)

Education

Ph.D., Astronomy, University of California, Riverside 2025 – expected Summer 2027

Advisors: Bahram Mobasher (UCR) and Shoubaneh Hemmati (Caltech/IPAC)

M.Sc., Astronomy, University of California, Riverside 2024 – 2025

M.Sc., Physics, McGill University, Montréal, Canada 2021 – 2023

Thesis: *Parameter Estimation of the Global 21 cm Signal*. Advisors: Jonathan Sievers and Oscar Hernández

B.Sc., Physics, Sharif University of Technology, Tehran, Iran 2018 – 2021

Publications

10 papers: 1 first-author (*ApJ*, under revision) and 9 co-authored (5 refereed, including *Nature Astronomy*). Four further first-author papers are in preparation. The full list is attached.

- **Haghjoo, A.**, Hemmati, S., Mobasher, B., et al. 2026, *Learning to See Sharper: A Physics-Informed Artificial Intelligence Framework for Super-Resolving Galaxy Spectra*, *ApJ*, under revision, arXiv:2603.18357
- **Haghjoo, A.**, Hemmati, S., Mobasher, B. *Sharper at What Cost? Benchmarking Deep Learning Against Classical Deconvolution for Galaxy Spectral Super-Resolution*, to be submitted to *ApJ*
- **Haghjoo, A.**, Hemmati, S., Mobasher, B., Chartab, N., de la Vega, A. *Recovering Emission Lines Without Inventing Them: Recoverability-Calibrated Spectral Super-Resolution for Roman*, to be submitted to *ApJ*
- **Haghjoo, A.**, Barish, B. *g21fit: A Unified Parameter-Estimation Package for the Global 21 cm Signal*, to be submitted to *ApJ*
- **Haghjoo, A.**, D'Aloisio, A. *Recovering the Ionized-Bubble Size Distribution from Stacked $\text{Ly}\alpha$ Spectra*, in prep.

Telescope Proposals (PI)

JWST Cycle 6 Archival Research (submitted), *Grating-Quality Emission Lines from the JWST Prism Archive* 2026

Research Experience

Graduate Researcher, University of California, Riverside Dec 2024 – present

- Physics-informed super-resolution of JWST/NIRSpec prism spectra ($R \sim 100 \rightarrow 1000$), trained on 2,858 JADES prism-grating pairs; line-token attention with a supervised presence gate (with B. Mobasher, S. Hemmati)
- Benchmark against seven classical deconvolution methods, quantifying false-line rates and amplitude bias
- Recoverability-weighted, anti-hallucination super-resolution and photometry-conditioned redshifts for the Roman grism survey, trained on 36,404 simulated OpenUniverse2024 spectra
- Ionized-bubble sizes from stacked $\text{Ly}\alpha$ damping wings with a CNN and simulation-based inference, applied to all public JWST/NIRSpec medium-resolution spectra at $z = 7-8$ (with A. D'Aloisio)
- Keck 2D spectral inspection and redshifts for the H2O catalog (with B. Mobasher)
- Co-author on COSMOS-Web and ALPINE-CRISTAL-JWST survey papers

Graduate Researcher, McGill University Sept 2021 – Dec 2023

- Parameter estimation for the global 21 cm signal with MCMC and Levenberg–Marquardt fits to ARES simulations; tests of non-standard physics against EDGES data (with J. Sievers, O. Hernández); extended since into *g21fit* with emulator-accelerated MCMC and neural posterior estimation (with B. Barish)

Undergraduate Researcher, Sharif University of Technology Jan 2020 – Sept 2020

- Machine-learning classification of OGLE variable stars (KNN, random forests, CNNs) under class imbalance (with S. Raeisi, S. Rahvar)

Open-Source Software

Released packages are pip-installable, documented and tested, with models and data on Hugging Face.

- `specsr` — super-resolution of JWST/NIRSpec prism spectra (doi:10.5281/zenodo.21943196)
- `specsrbench` — deep learning vs. classical deconvolution benchmark (doi:10.5281/zenodo.22104971)
- `specsr-roman` — calibrated super-resolution of Roman grism spectra, with public training set (doi:10.5281/zenodo.22119689)
- `zestimatr` — CNN spectroscopic redshifts with calibrated uncertainties (Code/Astro project)
- `g21fit` — global 21 cm parameter estimation: direct MCMC, emulator MCMC and neural posterior estimation (release in preparation)

Honors and Awards

UCR Dean's Fellowship	2024
McGill Faculty of Science Graduate Award	2023
Trottier Space Institute at McGill Graduate Fellowship; McGill Graduate Excellence Award	2021
Top 0.5% of 180,000 candidates, Iranian National University Entrance Examination	2017

Presentations

Invited

AstroAI Lunch Talk, Center for Astrophysics Harvard & Smithsonian	Apr 2026
Euclid US Tag-Up Meeting, Euclid Consortium	Mar 2026

Contributed

248th AAS Meeting, Pasadena (talk); PAI26 Conference on Physics and AI, Stanford (poster)	Jun 2026
Arizona Winter School: Cosmology in the Roman Era (poster)	Jan 2026
IR Spectroscopy from Space (Oct) and Greater IPAC Science Symposium (Aug), IPAC/Caltech (talks)	2025
AstroAI Workshop, CfA (talk); Cosmic Cartography with Roman, STScI (poster)	Jul 2025
Cosmology on Safari, South Africa (poster, Mar); Cosmology Seminar, Sharif (talk, Jun)	2023
McGill Radio Lab and Early Universe Cosmology group meetings (talks)	2022

Teaching, Mentoring, Outreach and Service

Teaching Assistant, Department of Physics and Astronomy, UC Riverside	2025 – present
Teaching Assistant, Department of Physics, McGill University	2022 – 2023
Teaching Assistant, Department of Physics, Sharif University of Technology	2019 – 2020
Co-organizer (with B. Mobasher), AI-in-education program for Riverside high-school teachers	2026 – present
Research mentor for senior students, STEM high school	Sept 2025 – Mar 2026
Representative, McGill Graduate Association of Physics Students; curation team, TEDxAveLorne	2023
Senior–freshman buddy program volunteer, Sharif University of Technology	2020 – 2021
Public observing nights, amateur astronomy outreach	2015 – 2020

Skills

Programming: Python (PyTorch, NumPy/SciPy, Astropy, emcee), Bash, C/C++, MATLAB, $\text{\LaTeX}$ · **Tools:** Git/GitHub, GitLab, SLURM, Weights & Biases, Hugging Face · **Languages:** Persian (native), English (fluent)

References

Bahram Mobasher

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Shoubaneh Hemmati

Research Scientist, Caltech/IPAC
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